



COMPUTER ARCHITECTURE

Lab # 2

Name: Horriya Noor

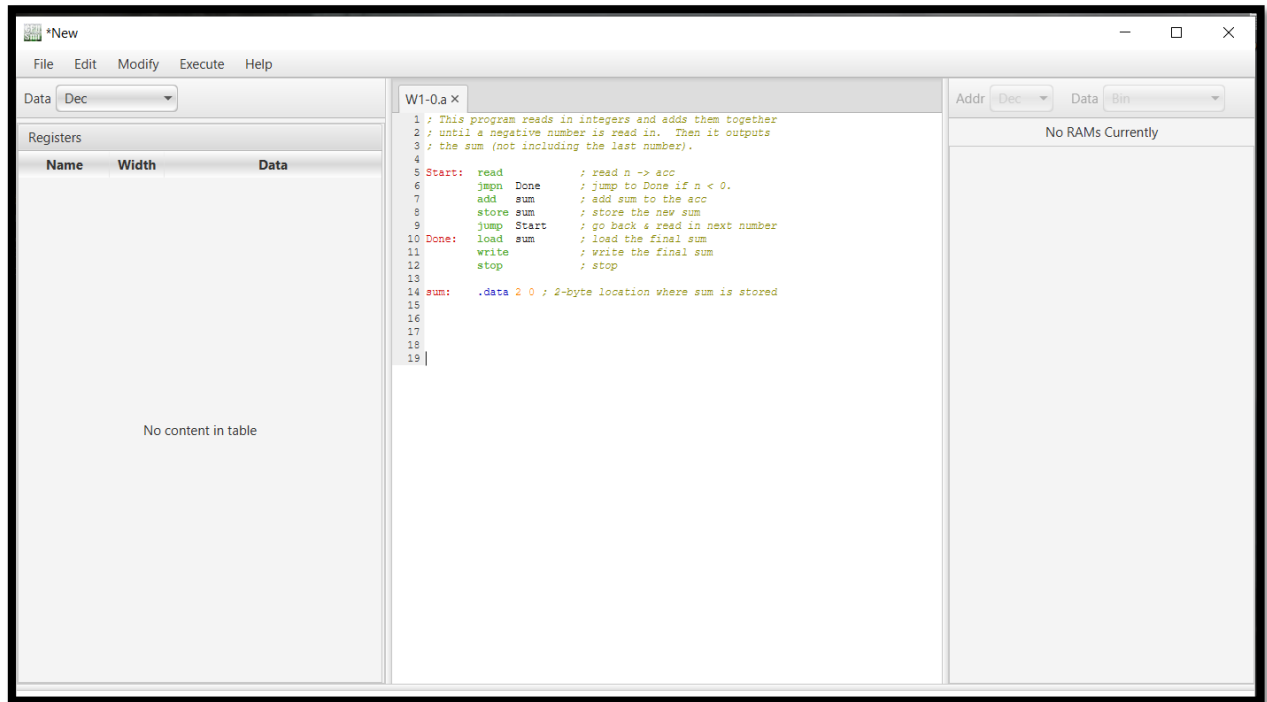
Sap ID: 64048

Department: BSCS

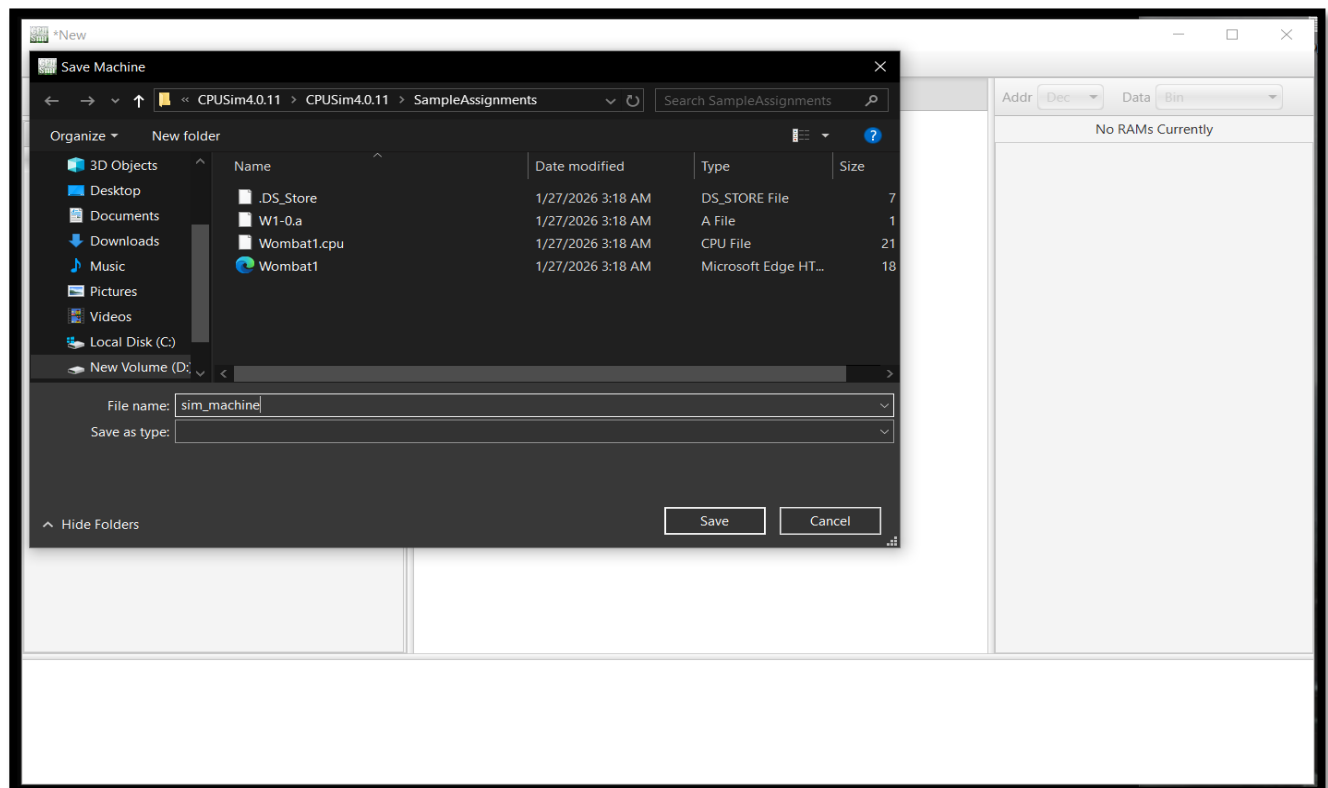
Semester: 4TH Semester

CREATING BASE MACHINE ON SIM CPU

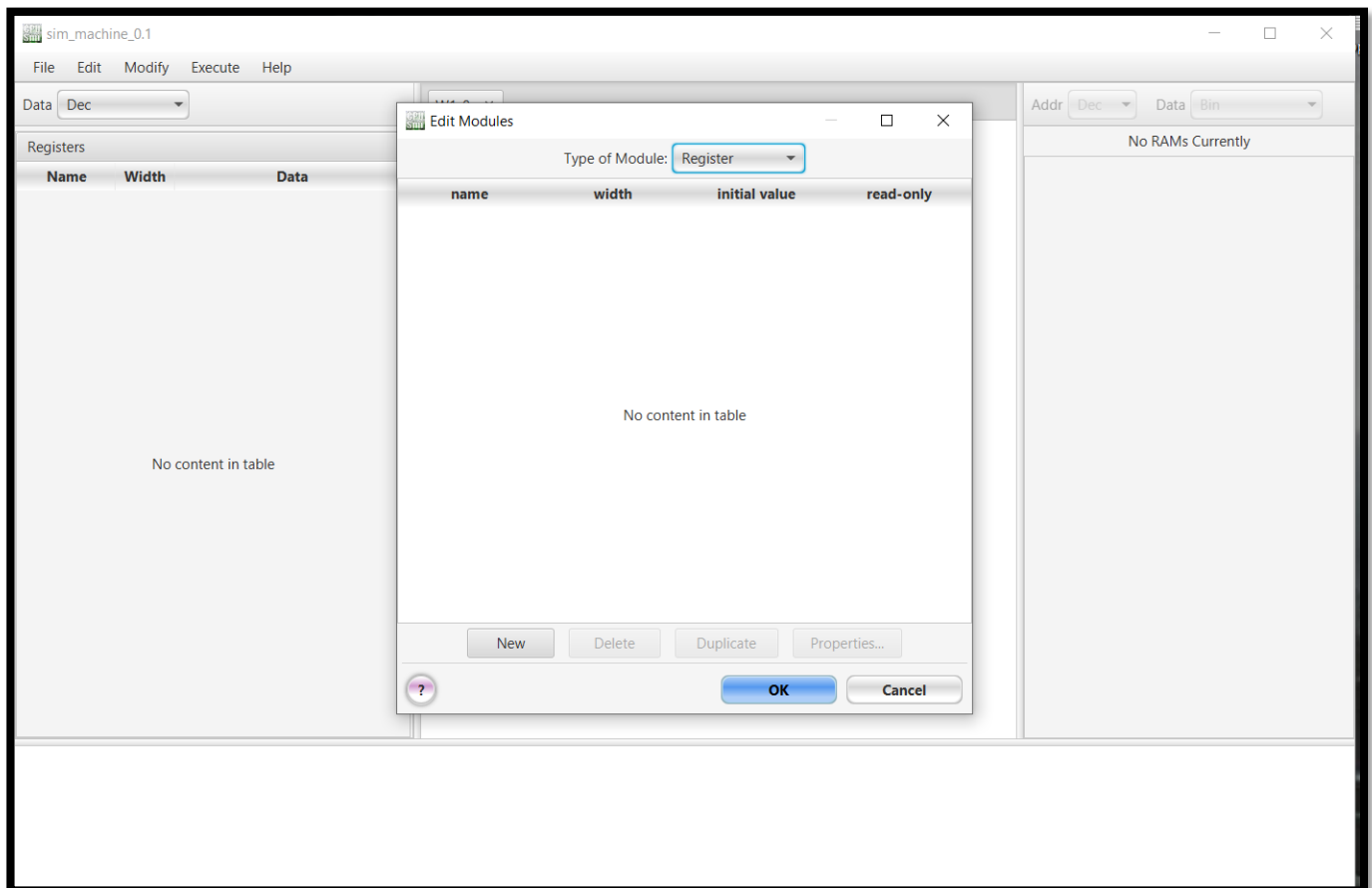
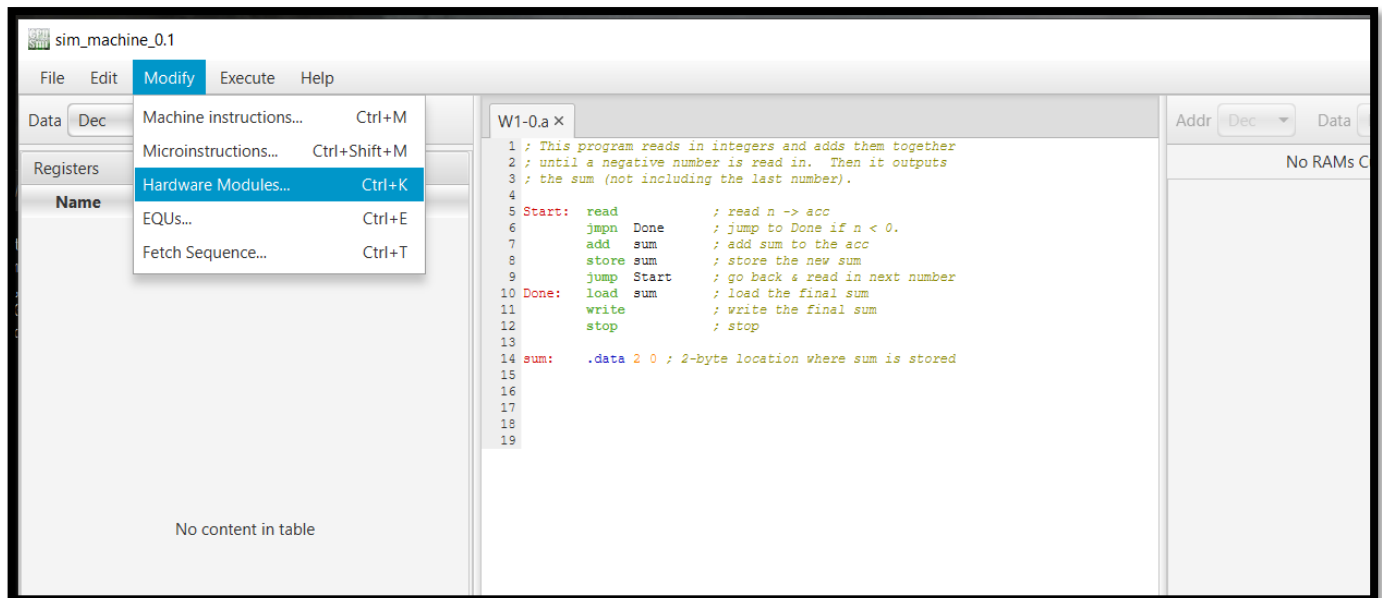
STEP 1: Opening CPU Sim.



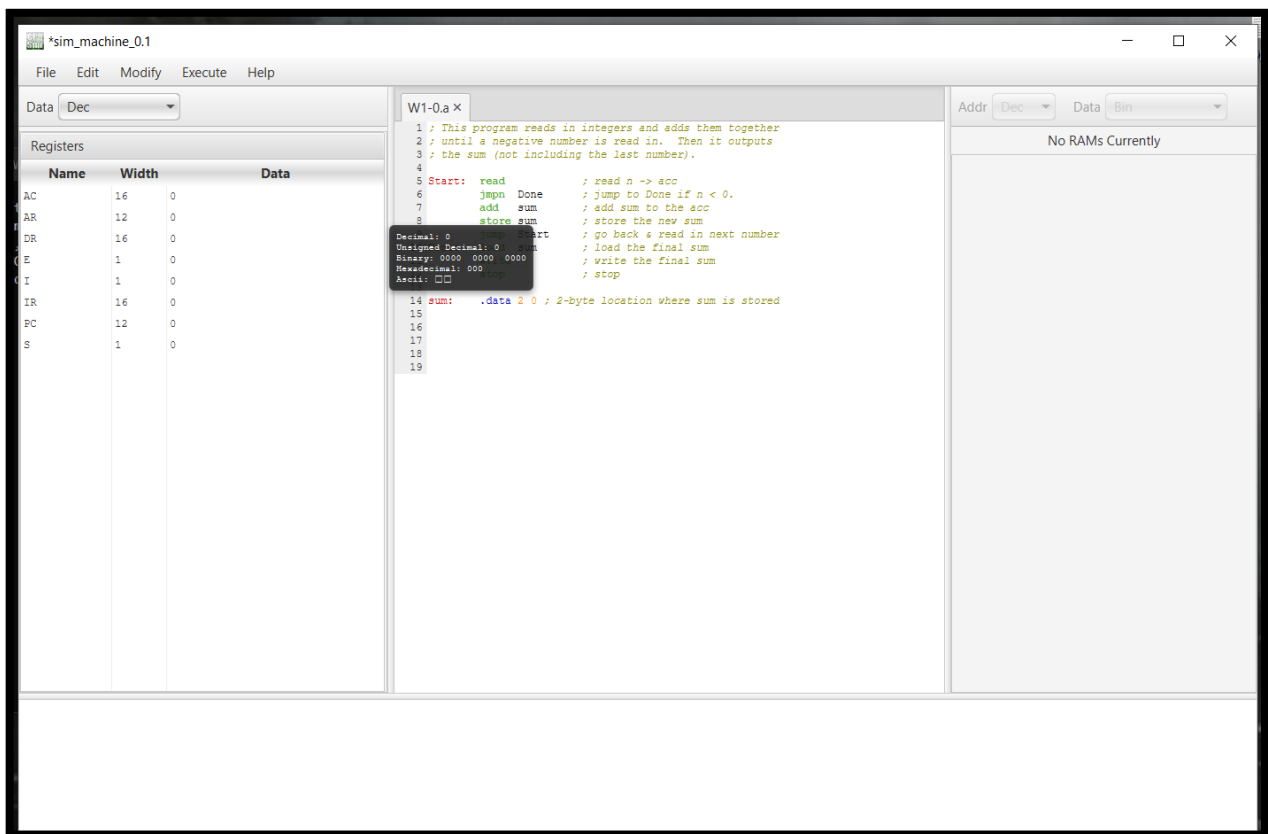
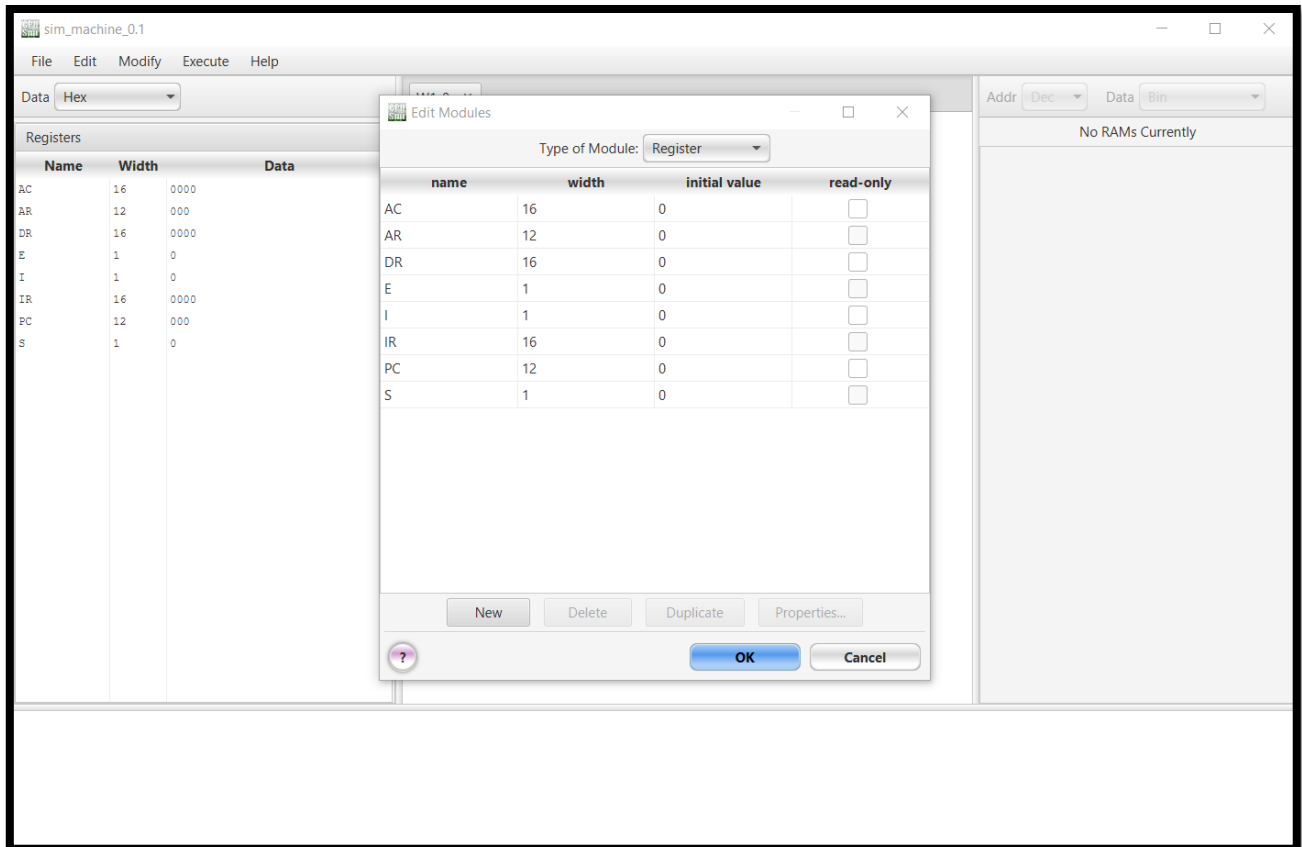
STEP 2: Naming the base machine.



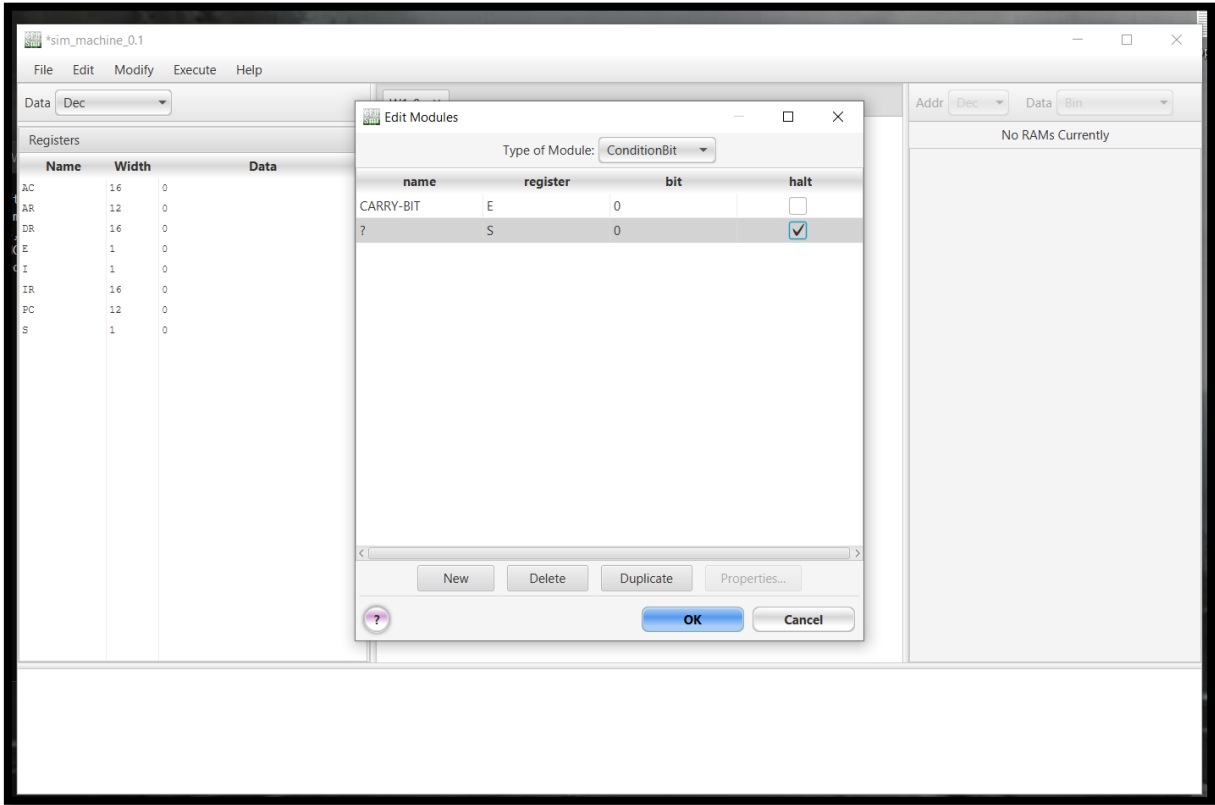
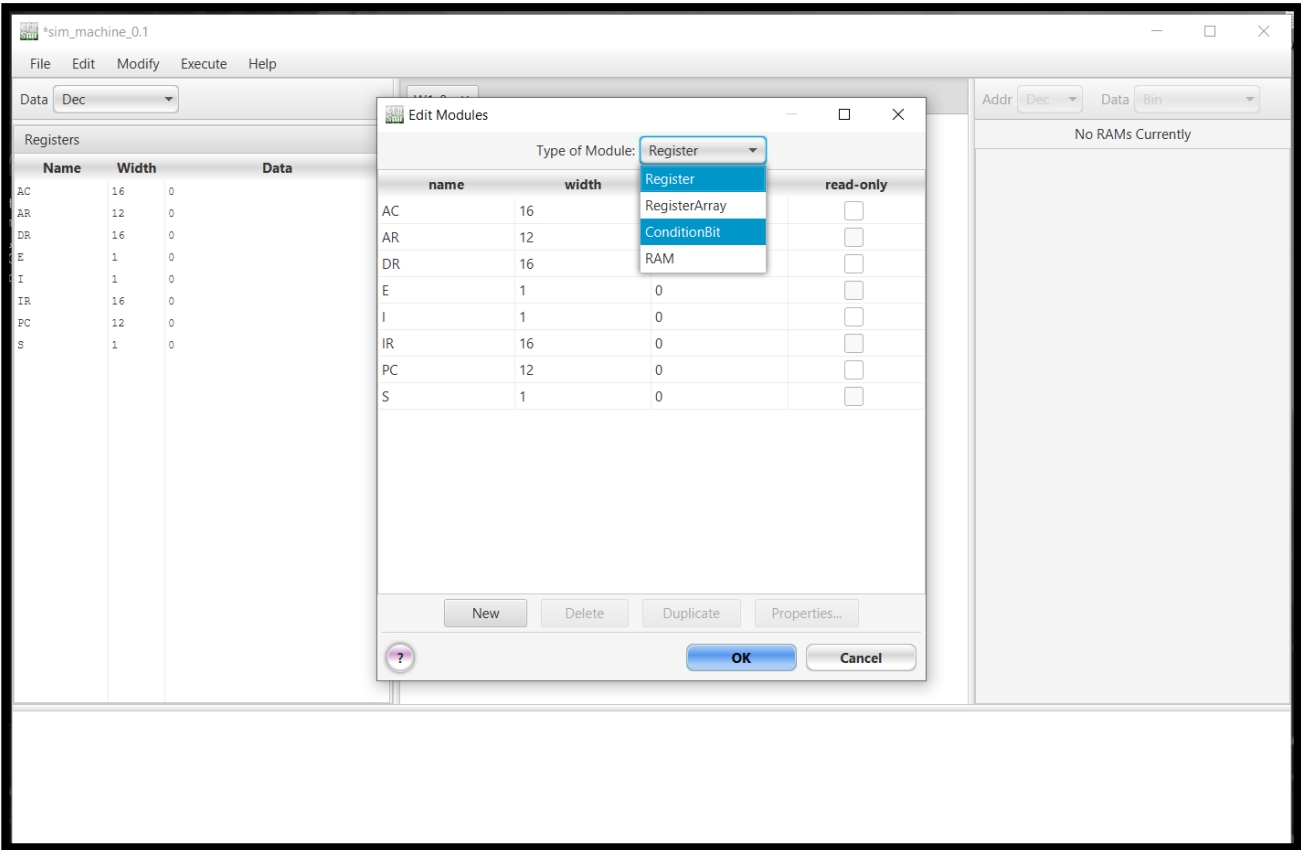
STEP 3: Go to modify , select **Hardware Modules**.



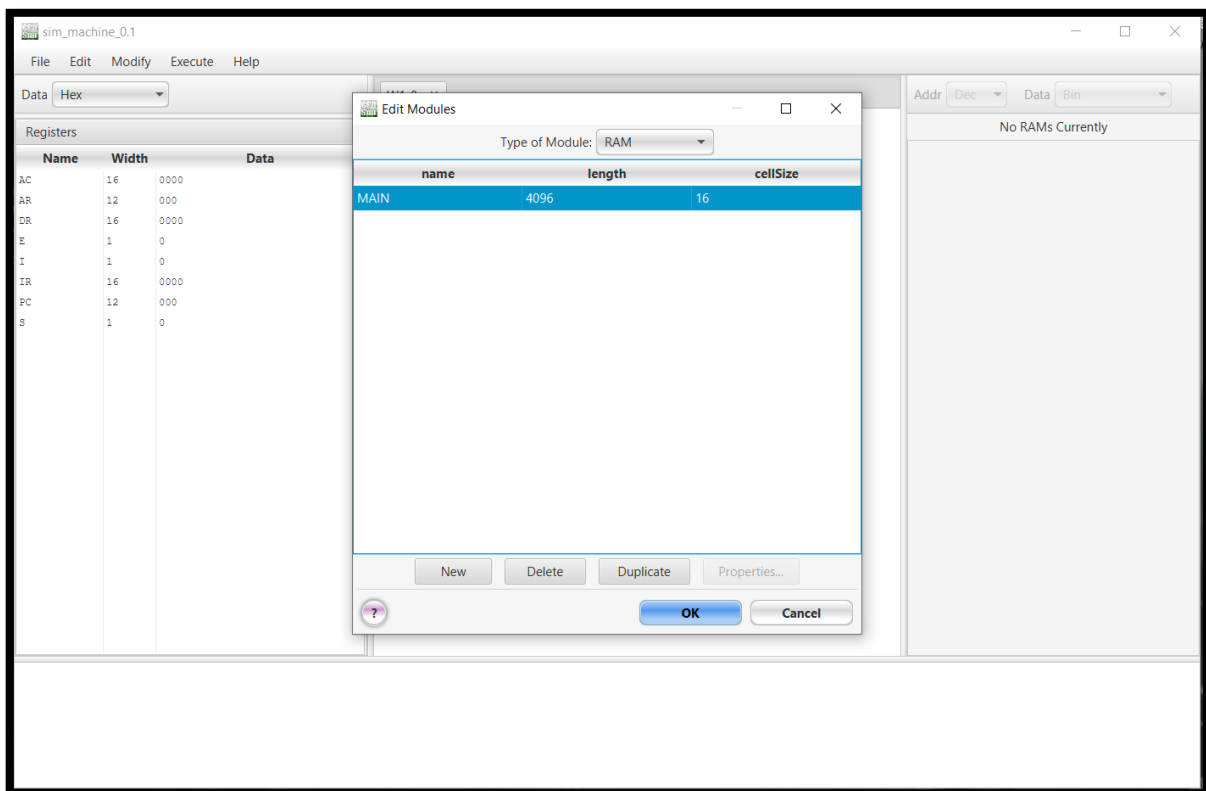
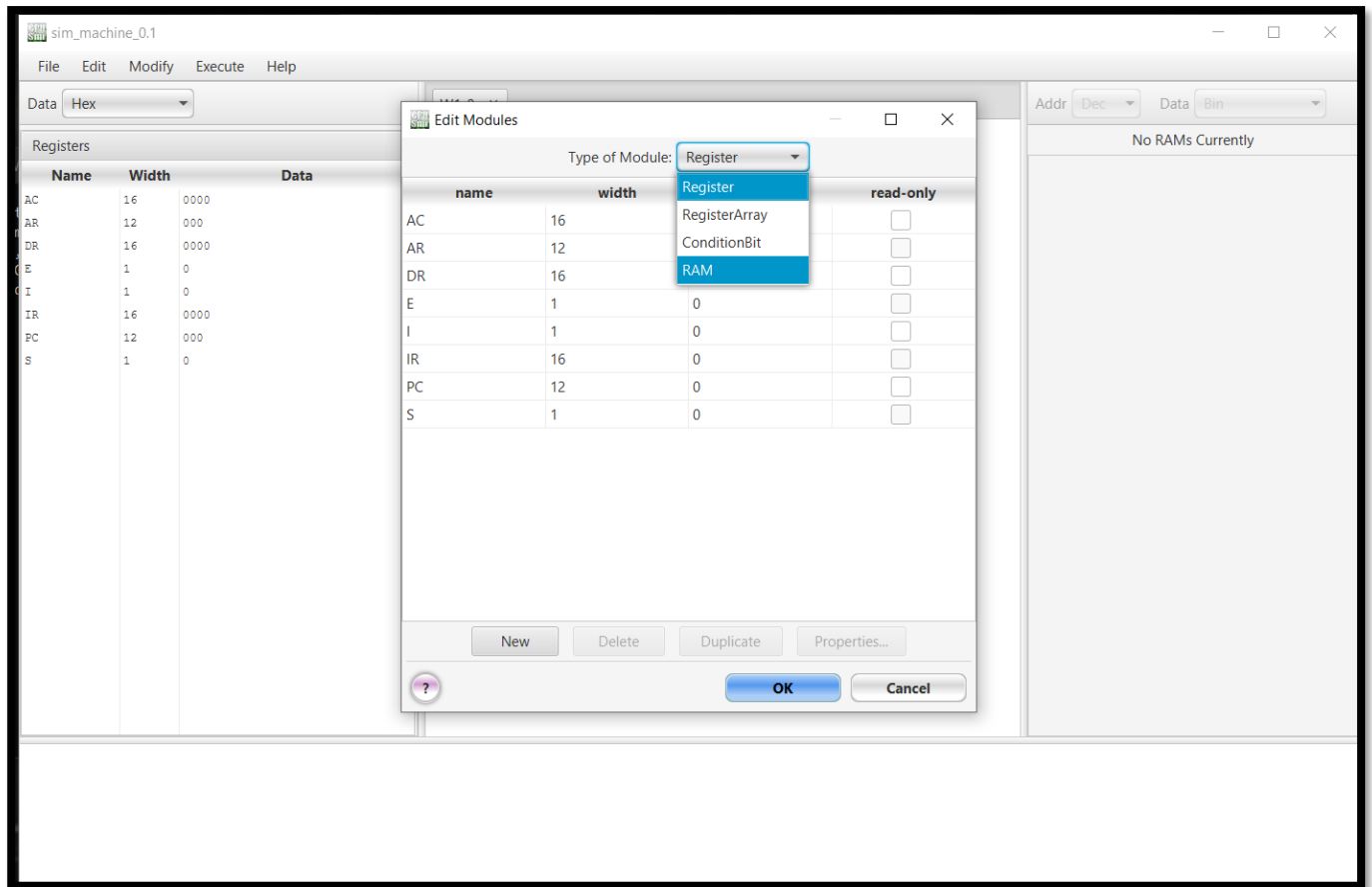
STEP 4: Adding Registers



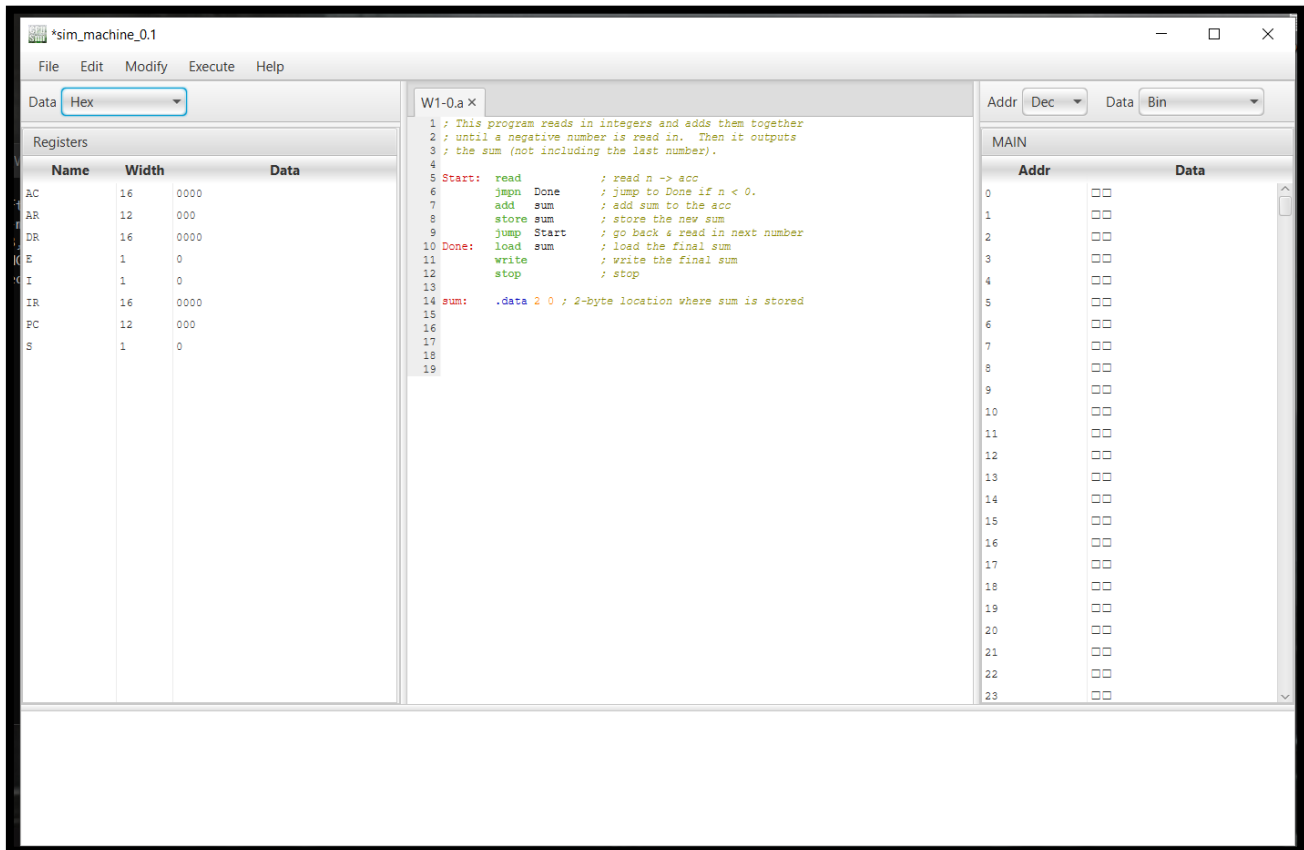
STEP 5: Adding Condition Registers



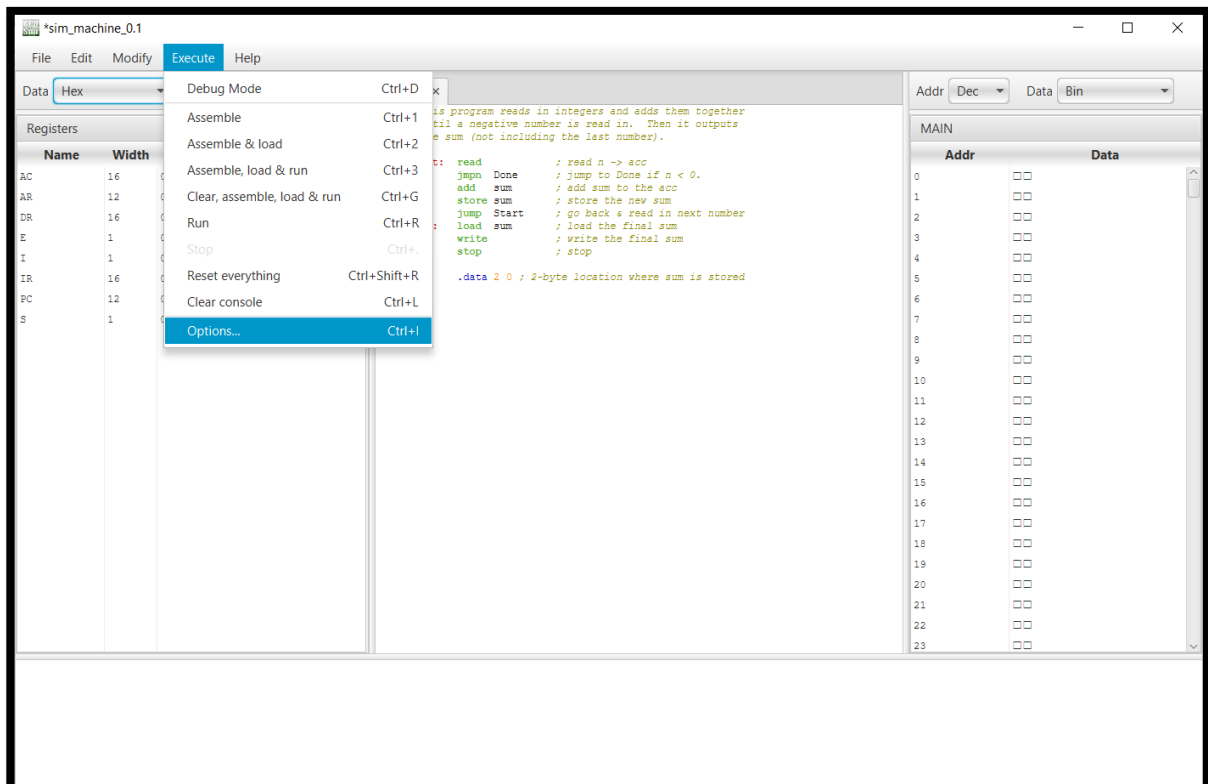
STEP 6: Adding RAM

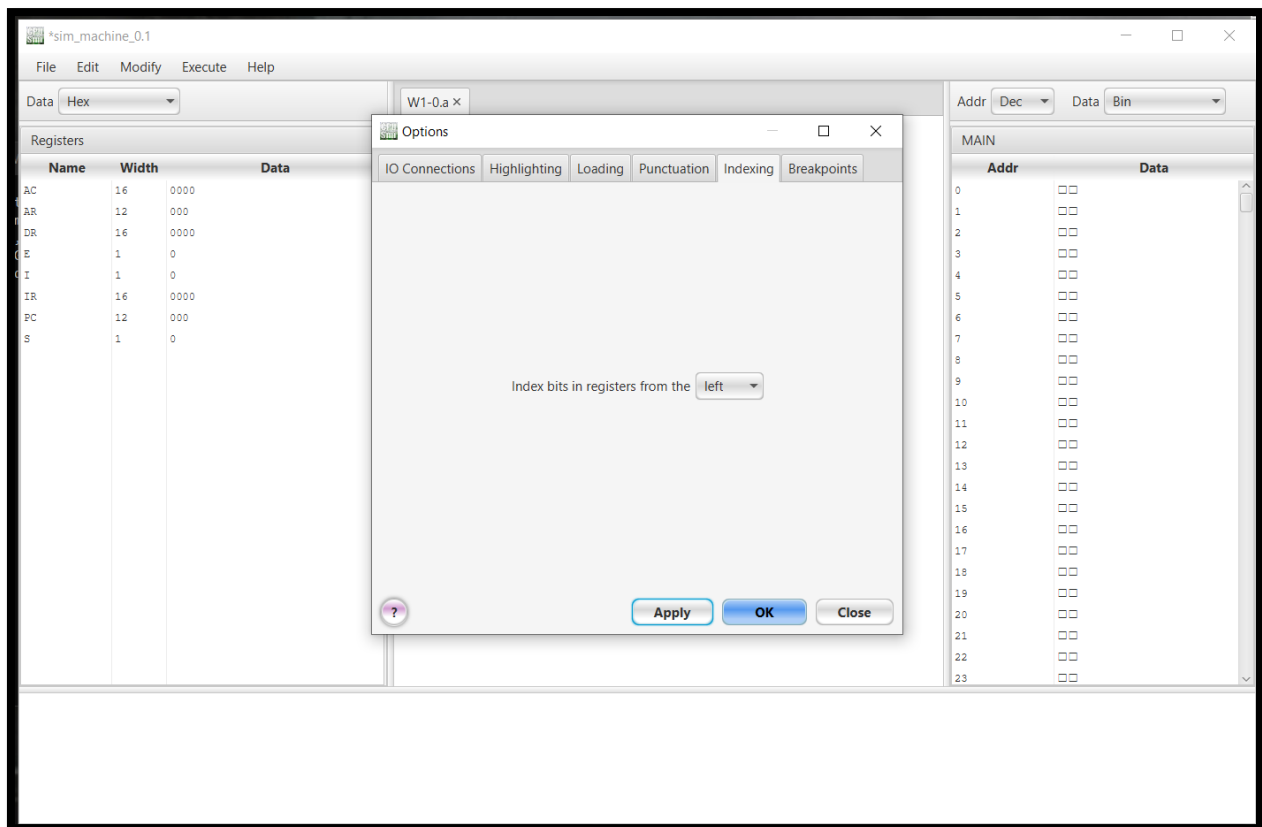


STEP 7: After changing Dec into Hex and adding RAM

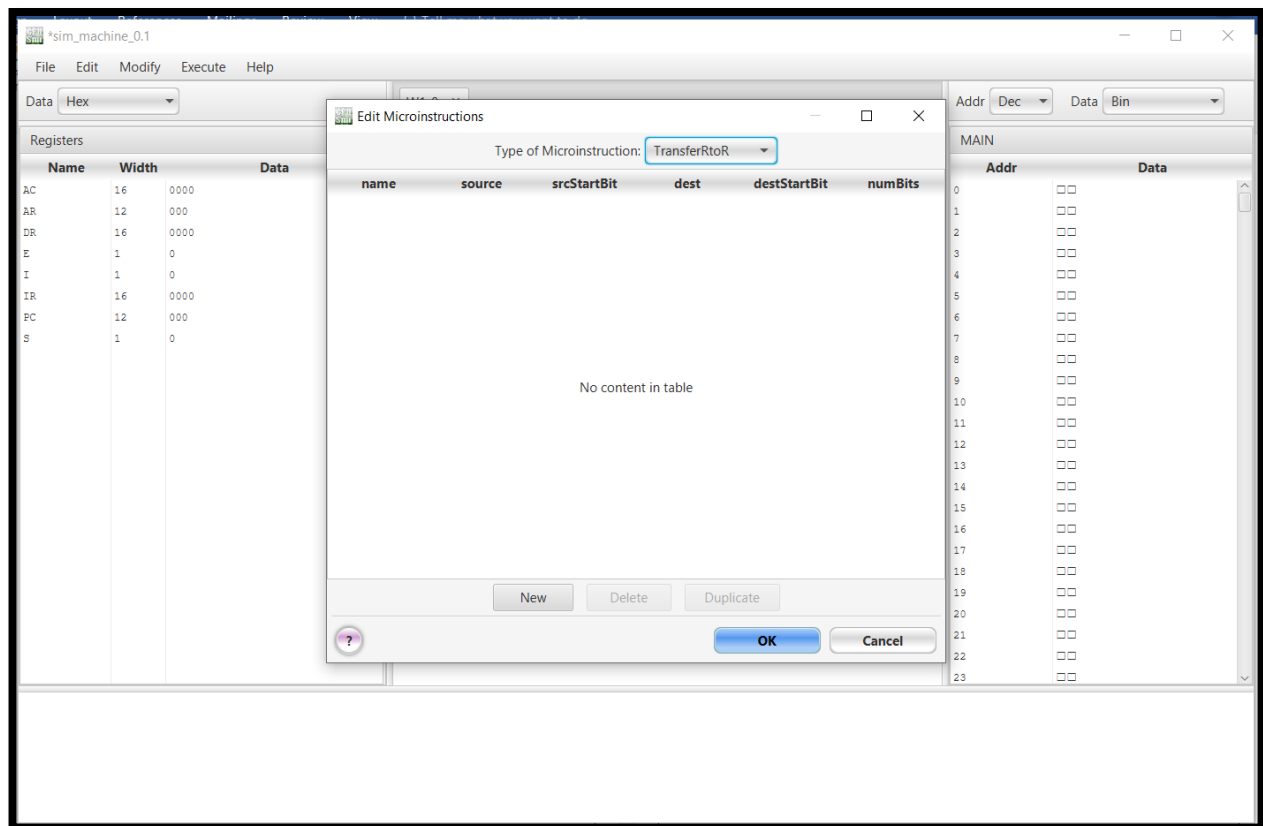


STEP 8: Selecting Option from top menu bar

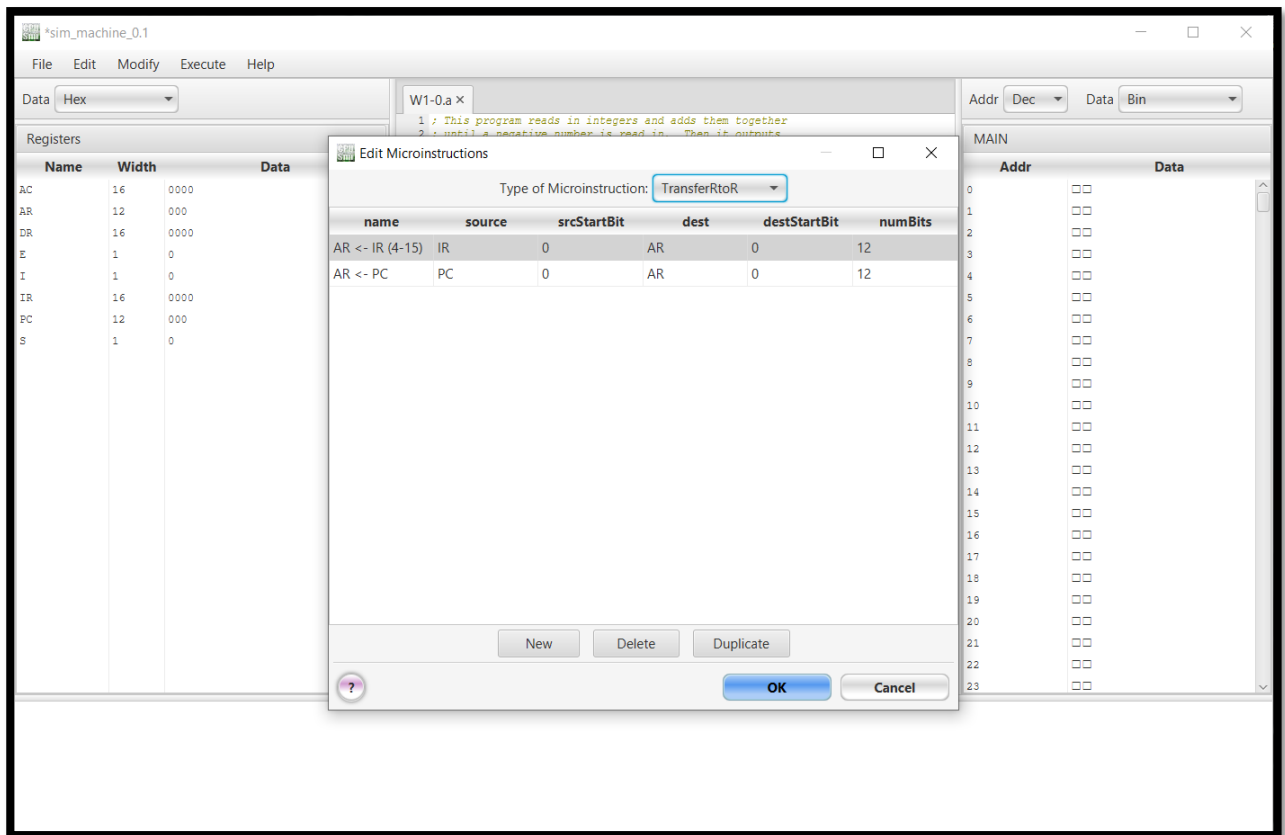




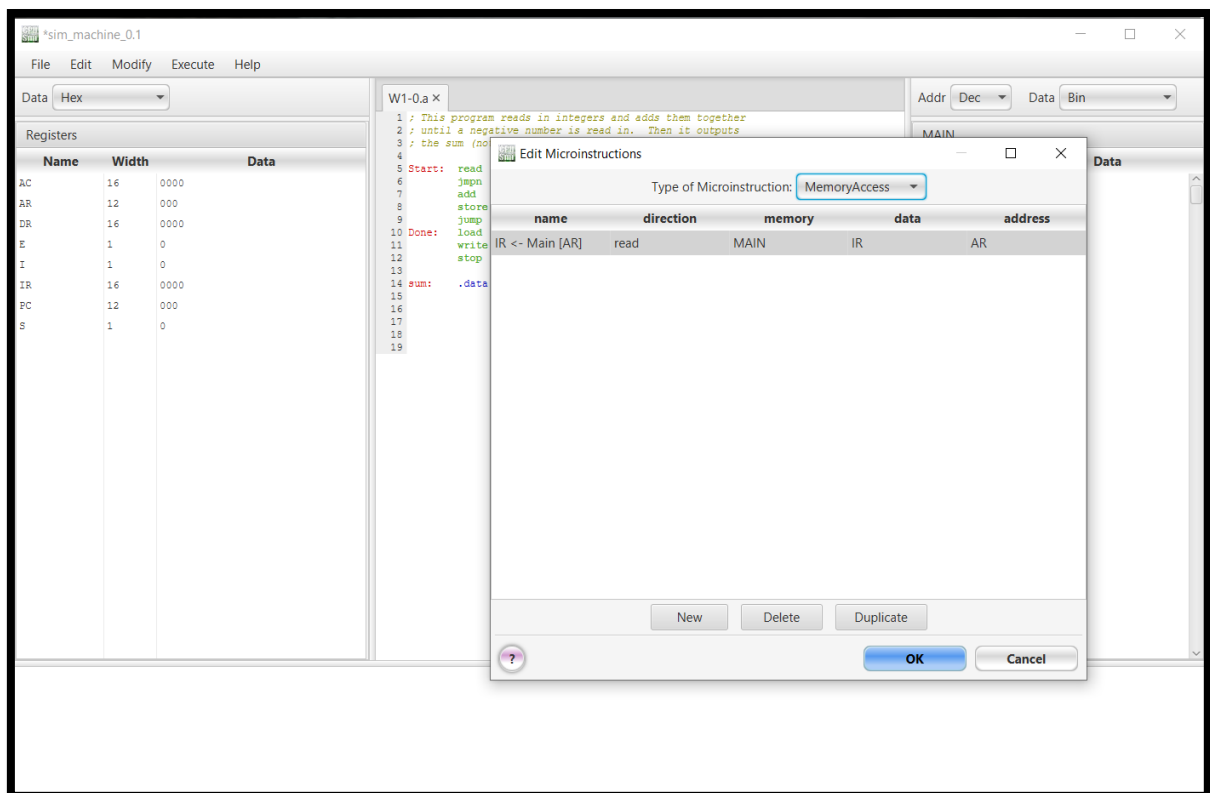
STEP 9: Adding the steps for Fetch Cycle



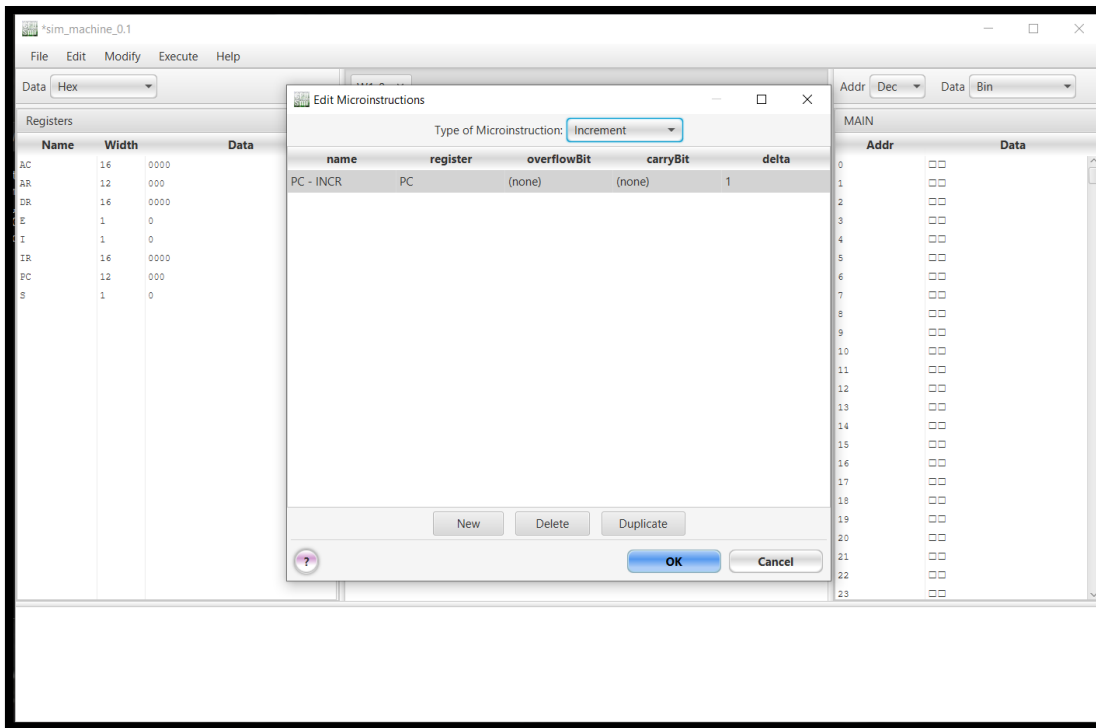
1. Transferring the address from PC to AR.



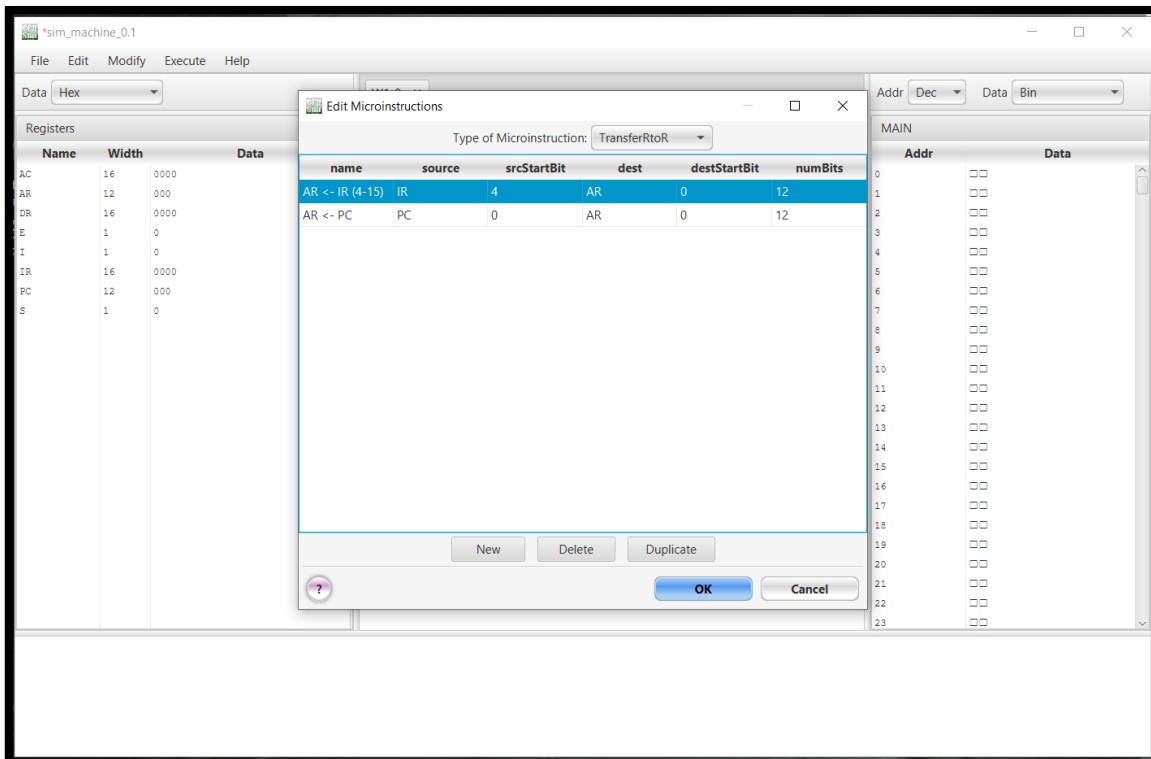
2. Reading the instruction from memory into IR.



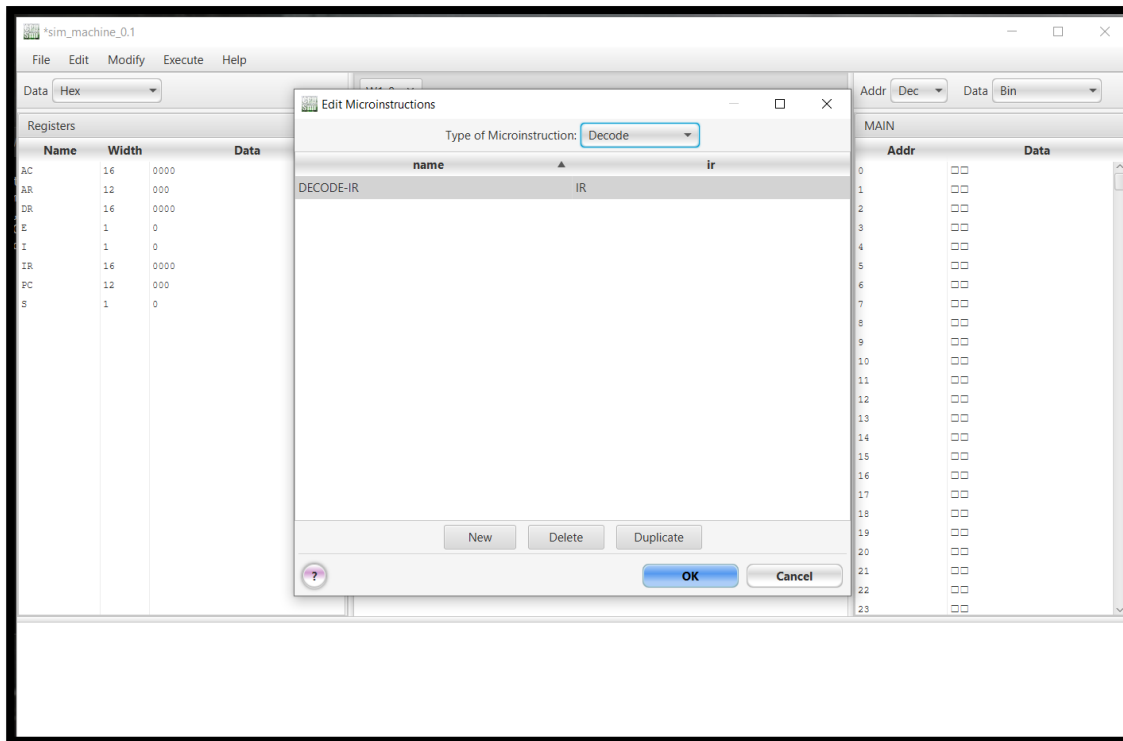
3. Increment PC to point to the next instruction.



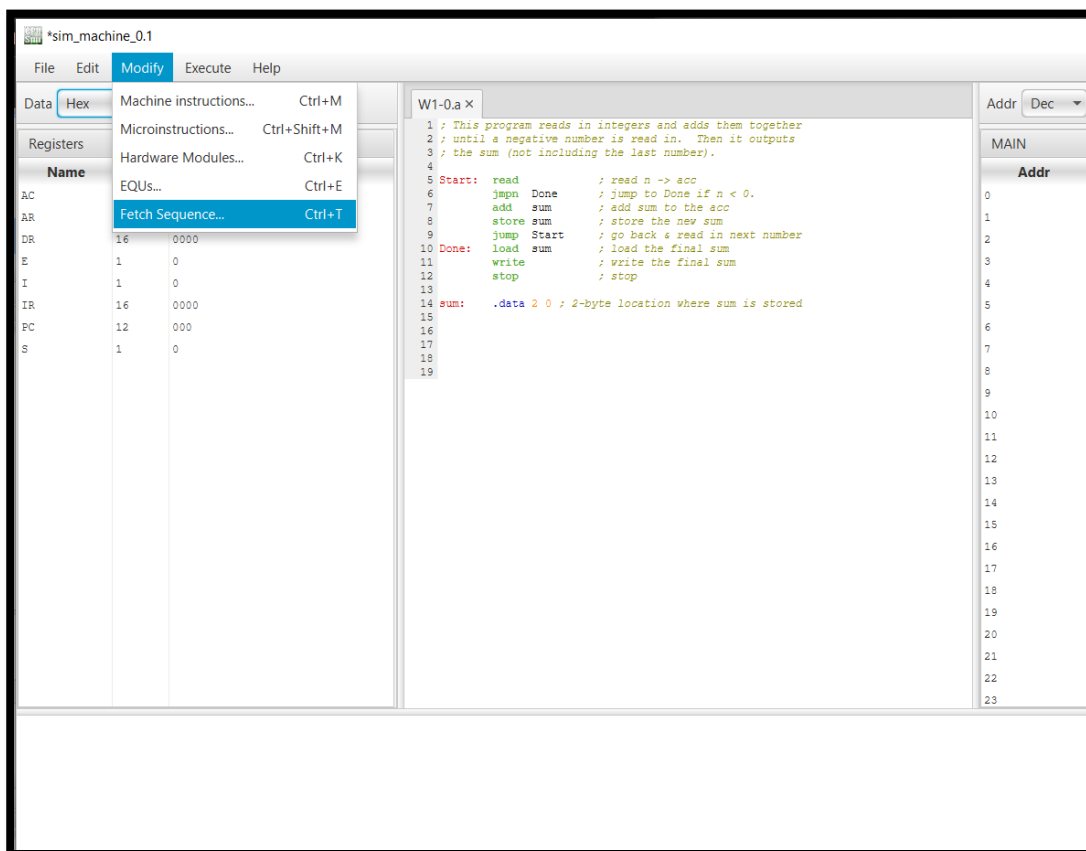
4. Extract the address part of the instruction and transfer it to AR.



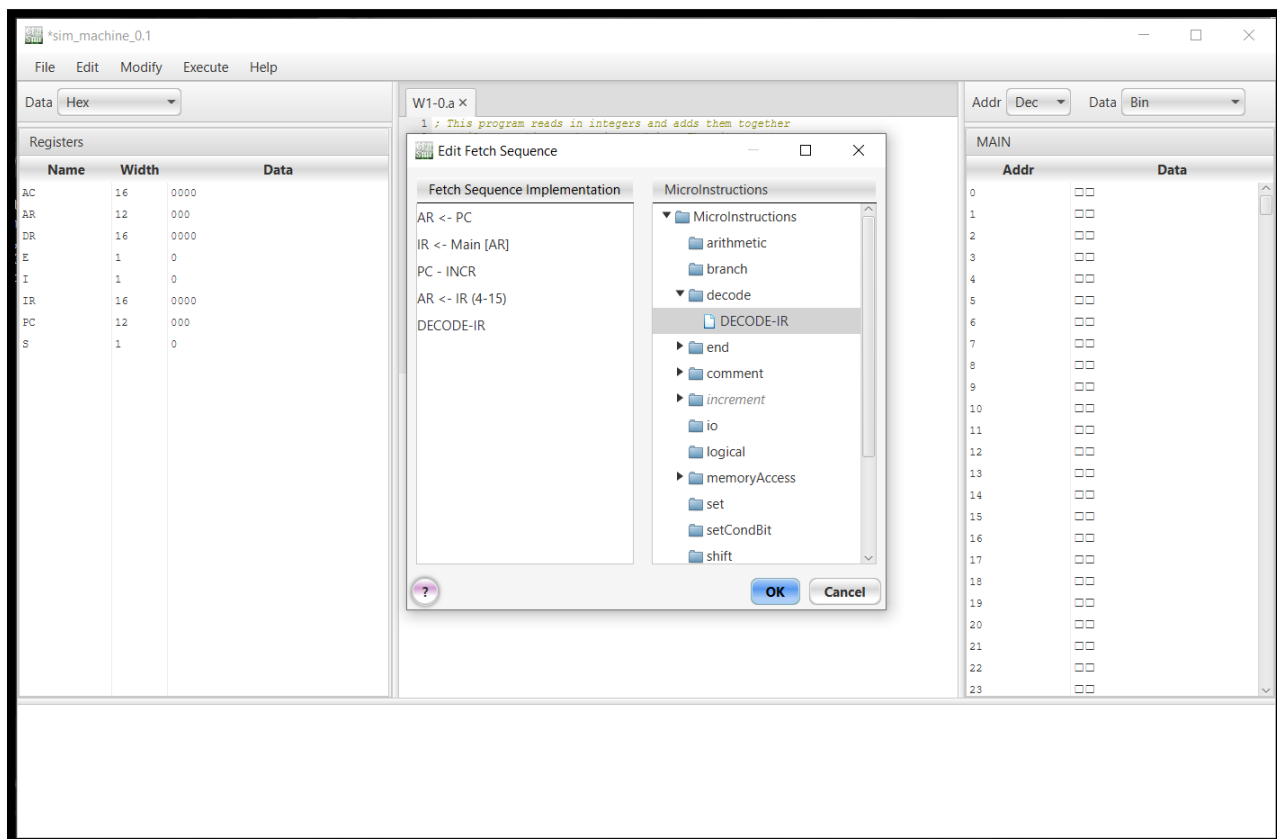
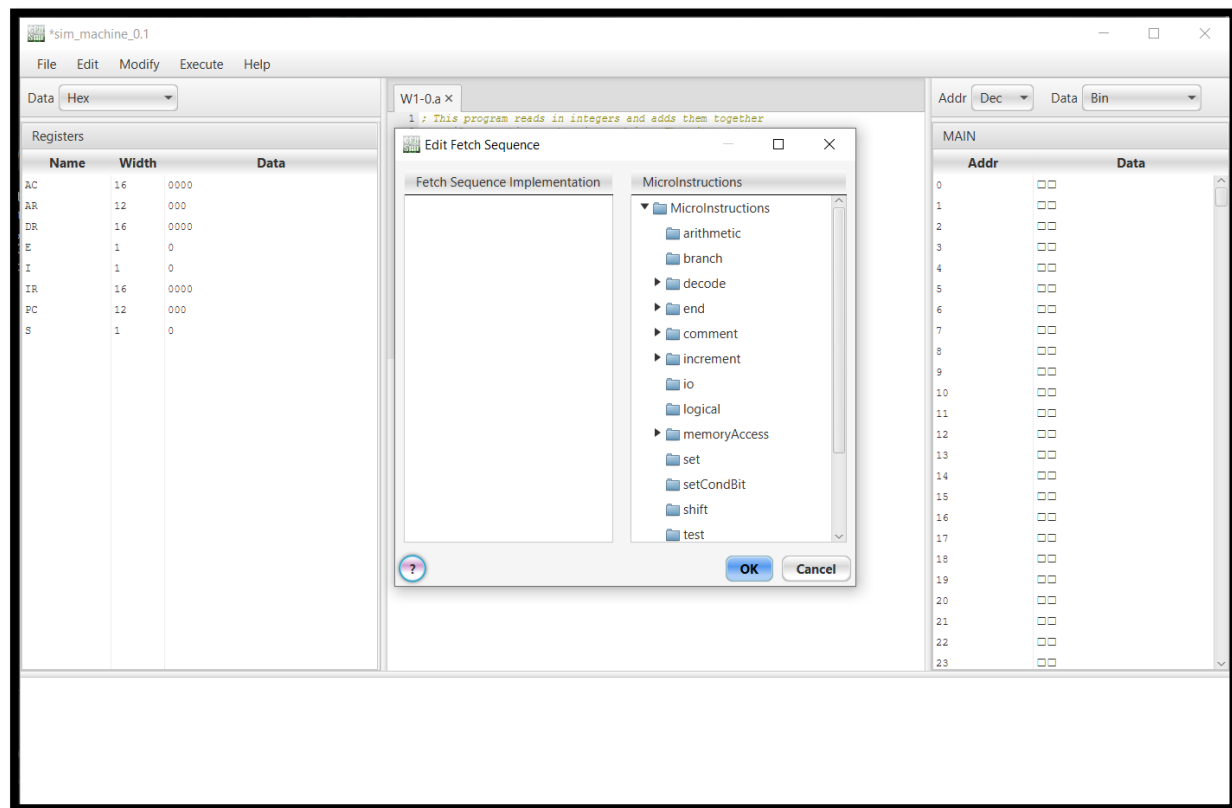
5. Decode the instruction for execution.



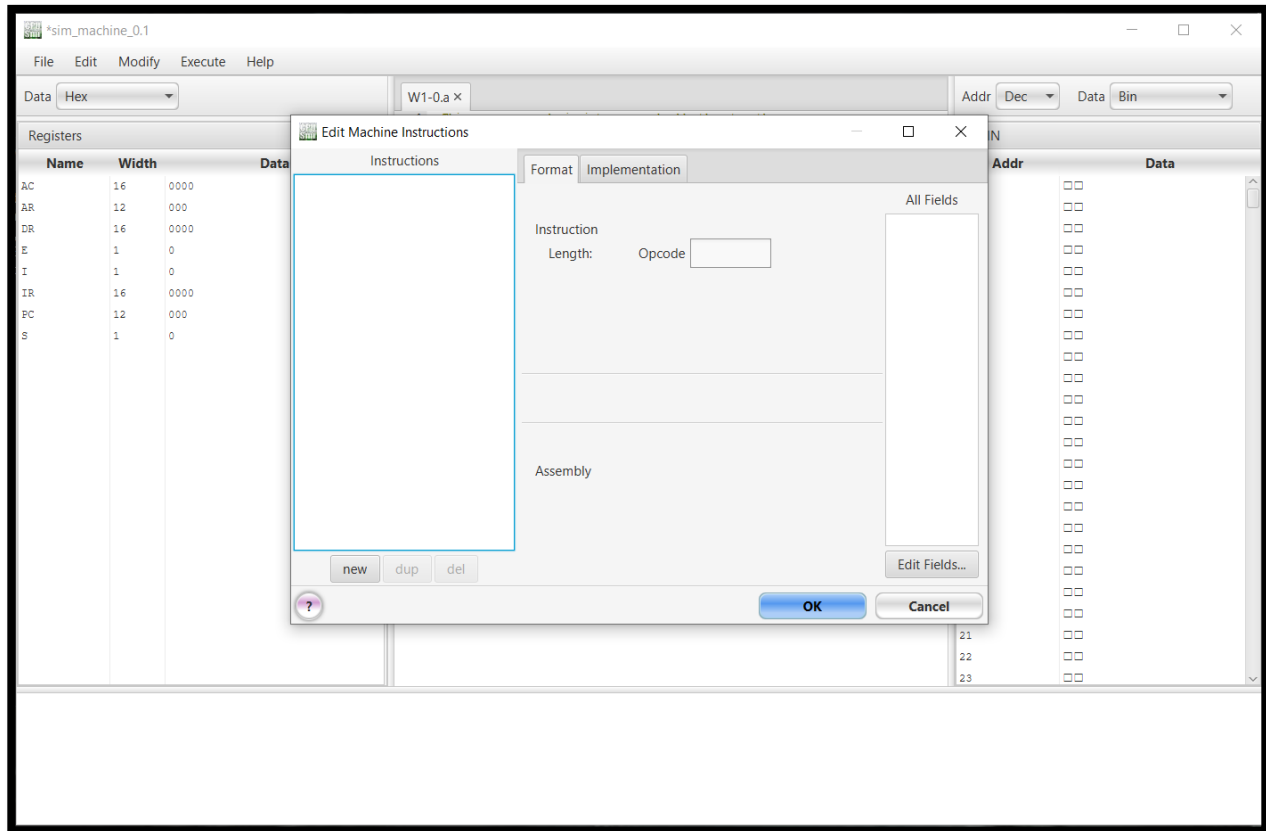
STEP 10: Selecting Fetch Sequence from top menu bar.



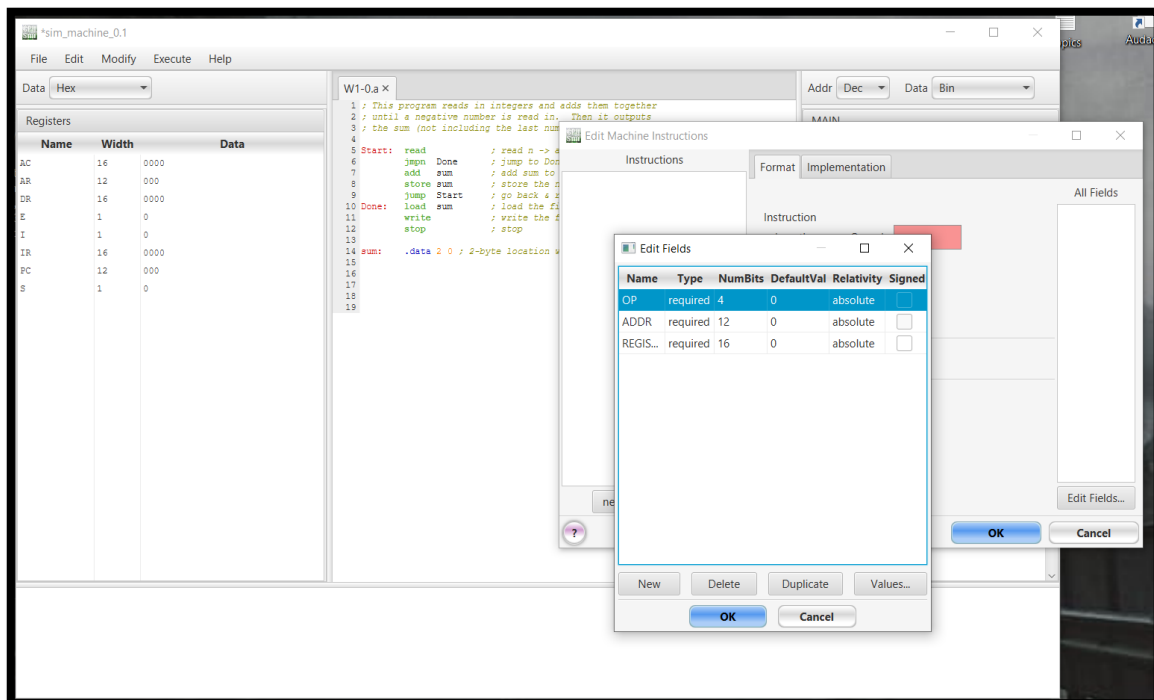
Fetching the instructions from microinstructions to Fetch Sequence Column.



STEP 11: Opening Machine Instructions from Top menu Bar.



Adding Fields in Machine Instructions .



STEP 12: The Setup for Base Machine is Complete.

